# CODE APEX

SKILL PASSPORT | VERIFIED SKILLS FROM REAL WORK

POWERED BY verzii

## TEAM DETAILS

- Team name: CodeCatalysts
- Team leader: Ved Shah
- Project: IM-03 Skill Passport: Verified Skills from Real Work

# Problem Statement & Target Need

## Who is affected

Students and early-career job seekers need credible ways to prove skills. Recruiters and college placement cells need clear, quick-to-check evidence.

## The current gap

Résumés are self-reported, while proof is scattered across repositories, project documents, portfolios and certificates. It is difficult to separate claims from demonstrated work.

## Why it matters now

Internship screening needs practical signals, not keyword lists. Generative AI can connect work to skills—but every claim must be checked against real sources.

## THE NEED

**A fast, transparent way to answer :**

**“What has this person actually demonstrated?”**

The trust gap is not a lack of skill lists. It is a lack of inspectable evidence behind each skill.

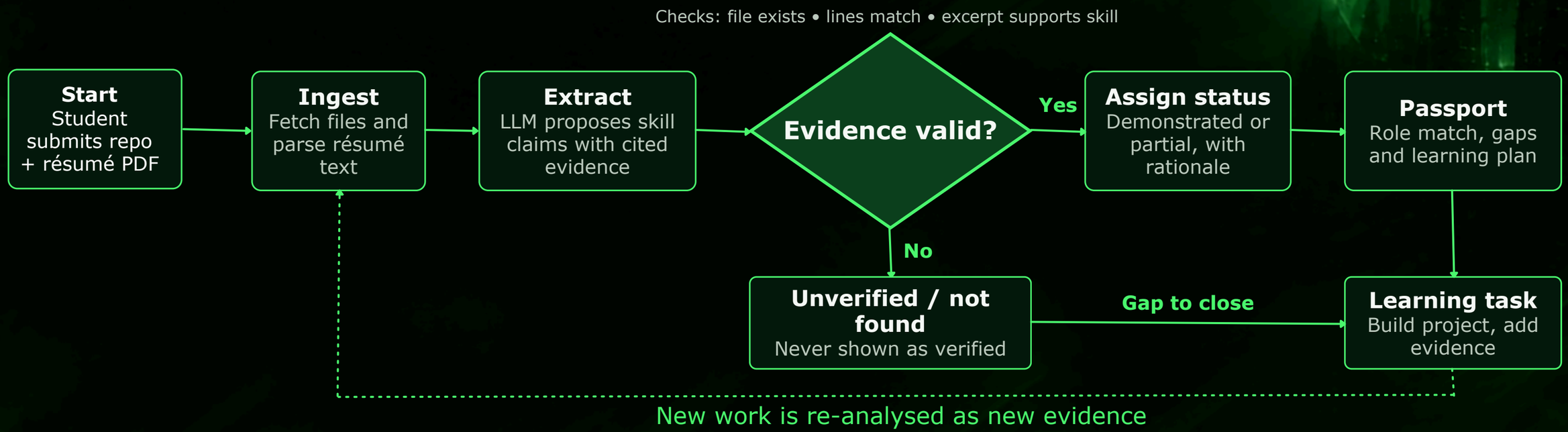
# Proposed solution & Innovation

**EVIDENCE BEFORE CLAIMS**
Evidence-first, not résumé-first: every demonstrated skill opens a source file, snippet or project artifact. Unsupported claims stay unverified.

**EXPLAINABLE ROLE FIT**
Show demonstrated, partial and missing requirements, not a mysterious score. The LLM only proposes claims; app code verifies the evidence exists and supports them.

**CLOSE THE LOOP**
Turn each gap into a practical task, then reassess new work as evidence.

## VERIFICATION FLOWCHART | CORE MECHANISM




**MVP feasibility:** one public GitHub repository • one résumé PDF • one LLM API • transparent matching rubric • sample data as live-demo fallback

# System Architecture & Tech Stack

## CLIENT LAYER
Next.js • React • TypeScript • Tailwind

**Landing & Submission**
Repo URL, résumé PDF, job description

**Student Dashboard**
Skill statuses and role coverage

**Passport & Evidence Viewer**
Source file, line range and excerpt

**Role Matcher & Learning Plan**
Requirement-level gaps and tasks

**Recruiter View (read-only)**
Shareable link with revocable access

HTTPS

JSON

## APPLICATION LAYER
Next.js Route Handlers • Zod validation

**Ingestion Service**
File fetch + authorship signals • résumé PDF extraction

**Skill Extraction (LLM)**
Structured JSON • Zod schema • prompt-injection guard

**Evidence Validation Gate**
App code checks file, lines and excerpt; unsupported claims are rejected

**Matching & Planning Engine**
Explainable rubric: role fit, gaps and learning tasks

**Access & Sharing Controls**
Auth, owner-controlled visibility, revocable share links

## DATA & EXTERNAL SERVICES
Supabase • OpenAI • GitHub

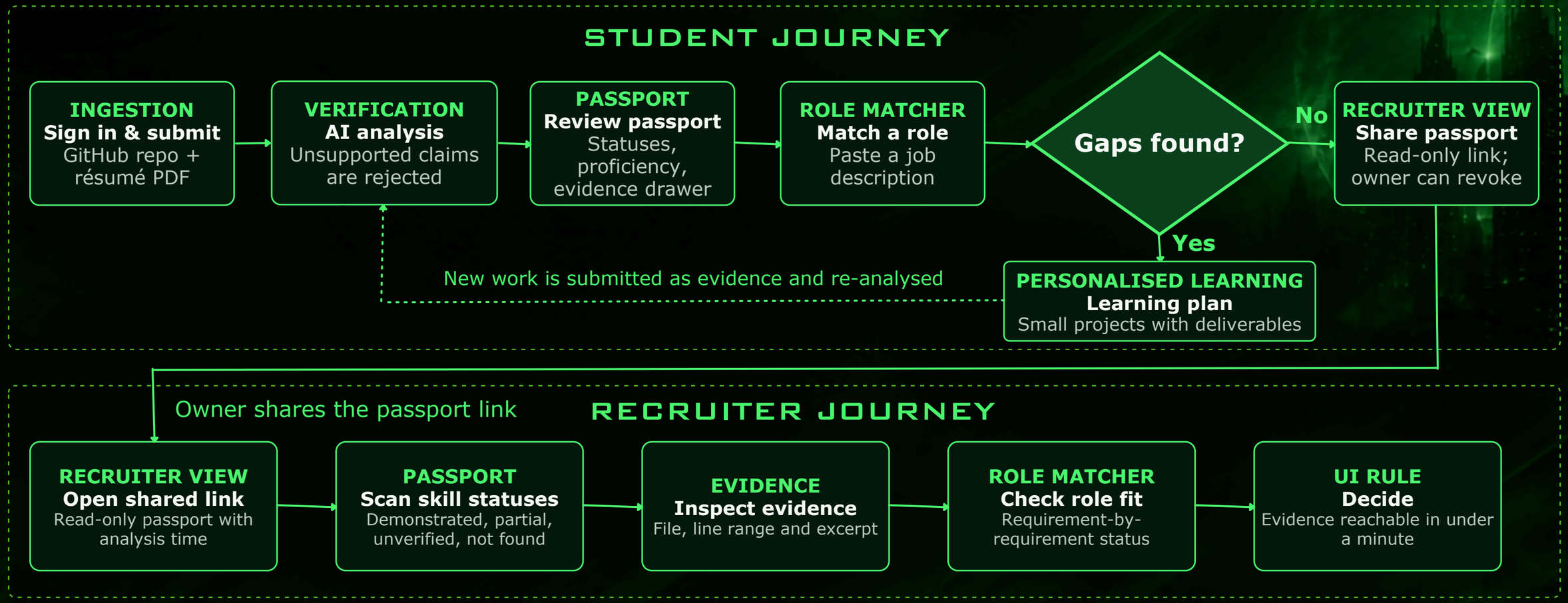
**GitHub REST API**
Repository tree and file contents

**Gemini API**
Structured skill extraction

**Supabase • PostgreSQL**
Auth and relational store
users • sources • source_fragments • skills • skill_claims • claim_evidence • role_assessments • role_requirements • learning_tasks • passport_shares

**Every accepted claim stores:** Skill • Status • Proficiency • Source • File/Line • Rationale • Last-Analysed
**MVP choice:** one **Next.js** codebase on **Vercel + Supabase**; add a **FastAPI** service only if **Python NLP/ML processing is required**.

# Key Features & UI/UX



# Implementation Roadmap & scope

How the MVP gets built, and what comes after the hackathon

### Foundation

Defining MVP scope, schema and skill taxonomy; Building the UI shell, Database and a Sample Repository.

### Ingestion

Fetching README and selected source files, parsing resume text and keep source locations.

### AI + Evidence

Extraction of skills with structured output; Implementing validation evidence in code and reject unsupported claims.

### Product Flows

Build passport, evidence viewer, role matcher, gap analysis and learning-plan view.

### Quality + Demo

Test reliability, security and responsive design, deploy and rehearse with sample data.

### Hackathon MVP

One public repo + one resume→ evidence-backed passport, traceable claims, job requirement gaps and a starter learning plan.

### Beyond the Hackathon

GitHub OAuth, multiple repositories, GitLab, certificate integrations, evidence freshness checks and placement-cell dashboards.

# Expected Impact & scalability

Portable, evidence-backed skills for students, recruiters and college placement cells.

### STUDENTS & JOB SEEKERS

Turn projects into a credible portfolio, understand skill gaps and focus learning on evidence they can build.

### RECRUITERS

Inspect proof behind a skill claim, compare applicants against a role and reduce dependence on keyword-only résumés.

### PLACEMENT CELLS

Spot cohort-level skill gaps and guide students toward practical projects and better-prepared applications.

## SCALABILITY PATH

### 1 MVP

One repo + one résumé; personal passport.

→

### 2 PILOT

Multiple sources and opt-in student cohorts.

→

### 3 SCALE

Integrations, source freshness and institution dashboards.

**SUCCESS MEASURES: 100% of "verified" claims have valid evidence references • Recruiter can inspect evidence within 60 seconds • Every job requirement receives an explicit status. These are design targets, not achieved results.**

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ARTIFICIAL INTELLIGENCE STUDENT FORUM

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# CODE APEX

BUILD BEFORE IT DOES.

**Presented by Team CodeCatalysts**
**Project: Skill Passport**

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# THANK YOU